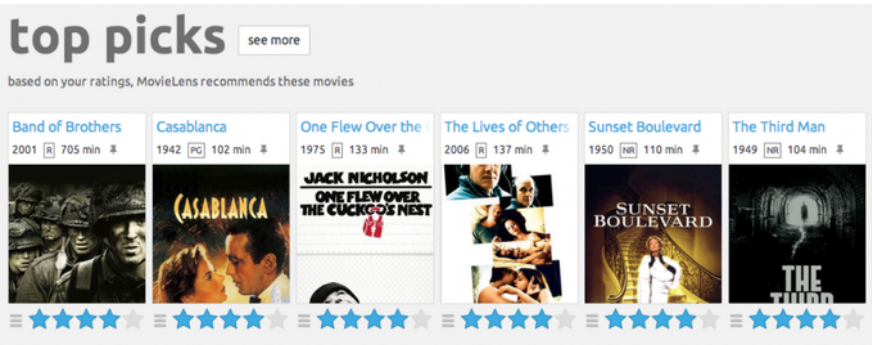


From Preferences to Predictions: Modeling Unseen Movie Ratings with Three Learning Approaches

Yujie Chen, Xinwen Zhang, Zehao Shi

Purpose



This project studies the rating prediction problem in recommender systems by predicting user ratings for unseen movies. Using the same MovieLens dataset, we compare SVD, Neural CF, and Bayesian PMF to evaluate how different modeling approaches affect predictive performance.

Background

Accurate rating prediction is a central problem in recommender systems because it helps platforms infer user preferences for items they have not yet rated. In the movie domain, this improves personalization by recommending films that better match individual tastes rather than relying only on overall popularity. We study this problem using the MovieLens dataset, a standard benchmark containing 1,000,209 ratings from 6,040 users on about 3,900 movies.

Methods

Shared setup part

- same MovieLens dataset
- same 80/10/10 train-validation-test split
- same input setting: user_id, movie_id
- same evaluation metric: RMSE
- predictions clipped to the original 1–5 scale

SVD-based Matrix Factorization

- low-dimensional latent factors for users and movies
- includes global mean, user bias, and movie bias
- predicts ratings through latent factor interaction

Neural Collaborative Filtering

- learns user and movie embeddings
- replaces linear interaction with an MLP
- captures nonlinear user–item relationships

Bayesian PMF

- probabilistic extension of matrix factorization
- models latent user/movie factors under Bayesian assumptions
- supports uncertainty-aware rating interpretation

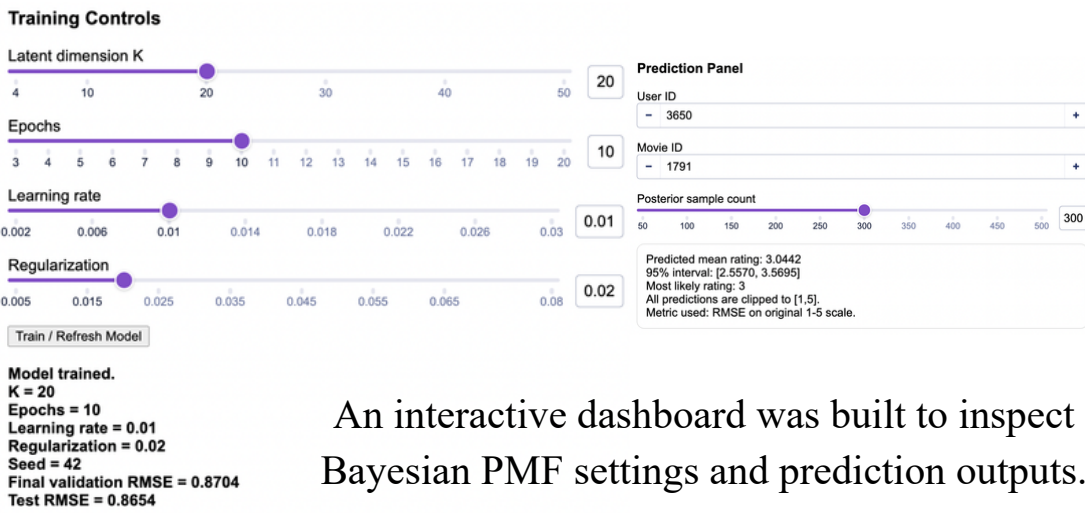
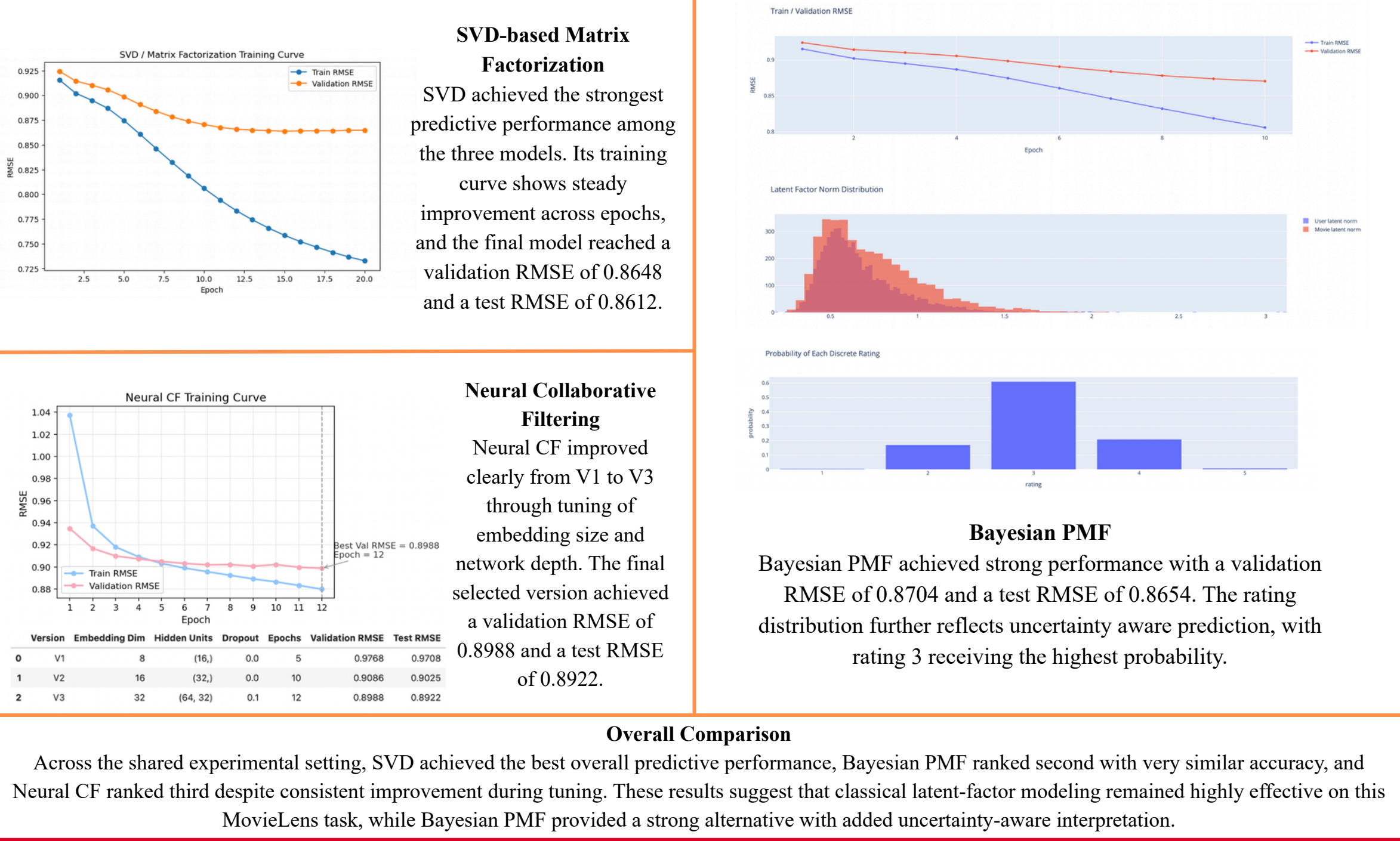
Conclusions

SVD achieved the best overall performance under the shared experimental setting. Bayesian PMF ranked second with very similar accuracy and added uncertainty aware prediction. Neural CF improved substantially from V1 to V3, but its final RMSE remained higher than the other two models. Overall, the results suggest that classical latent factor modeling remains highly effective for movie rating prediction when only user id and movie id are used as inputs.

Next steps

Future work could include richer user and movie features, stronger hybrid models, and more advanced Bayesian inference. These extensions may improve both prediction accuracy and interpretability.

Results



An interactive dashboard was built to inspect Bayesian PMF settings and prediction outputs.

References

Resnick, P., Iacovou, N., Suchak, M., Bergstrom, P., & Riedl, J. (1994). GroupLens: An open architecture for collaborative filtering of Netnews. In Proceedings of the 1994 ACM Conference on Computer Supported Cooperative Work (pp. 175–186).

Bell, R. M., & Koren, Y. (2007). Lessons from the Netflix Prize Challenge. ACM SIGKDD Explorations Newsletter, 9(2), 75–79.

Koren, Y., Bell, R., & Volinsky, C. (2009). Matrix factorization techniques for recommender systems. Computer, 42(8), 30–37.

He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T.-S. (2017). Neural collaborative filtering. In Proceedings of the 26th International Conference on World Wide Web (pp. 173–182).

Salakhutdinov, R., & Mnih, A. (2008). Probabilistic matrix factorization. In Advances in Neural Information Processing Systems, 20, 1257–1264.